

Quiz 2

Q1. Refer to the table below to answer the following questions:

Doctors			
DoctorID	DoctorName	DoctorFloor	DoctorDept
D01	House	1	Pathology
D23	Jack	2	Cardiology
D43	Doofenshmirt	3	Neurology

Patients		
PatientID	PatientName	AssignedTo
P077	Alice	D01
P188	Bob	D23
P195	Rick	D01

- Insert <null, "Melfy", 'A', "Psychology"> into the Doctors table. Identify all constraints which will be violated by this operation and explain how you will solve it. **[2 Marks]**
- Insert <"P077", "Yorick", "D21"> into the Patients table. Identify all constraints which will be violated by this operation and explain how you will solve it. **[2 Marks]**

- Ans: Entity Integrity Constraints, for null, solve: set as PK,
Domain Constraint, for "A", set proper datatype
- Ans: Key Constraint, for P077, set as PK
FK Constraint / Referential Integrity Constraint for D21, set proper FK and PK

Q2. You are tasked with designing a Pilot and Plane Management System. The system must meet the following requirements:

- Pilots will have a unique PilotID, full name, email, and license number.
- The PilotID is composed of pilotSchool and pilotNumber.
- Each plane has a unique PlaneID, plane model, capacity, and manufacturer.
- Each pilot can operate many planes, but each plane can only be operated by one pilot at a time.
- There are only 2 types of Pilots: Captain and Copilot. A pilot can not be both at the same time. Captains have flight_hour and Copilot have service_years.
- Planes are categorized by their type, such as Commercial, Cargo, or Private. A plane can be more than one type from these categories.
- Copilots can fly private planes. One copilot can fly one private plane at a time. A private plane can have only one copilot.

Now, draw an EER (Extended Entity Relationship) diagram of this system. **[6 Marks]**

Write down all assumptions (if any).

Solution:

